COURSE TITLE	**:**	**DATA COMMUNICATION AND COMPUTER NETWORKS**
COURSE CODE	**:**	**4262**
COURSE CATEGORY	**:**	**A**
PERIODS/WEEK	**:**	**4**
PERIODS/SEMESTER	**:**	**60**
CREDITS	**:**	**4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Physical Layer and Media	**15**
2	Data Link Layer	**15**
3	Network Layer	**15**
4	Transport and Application Layers	**15**

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	**1**	Understand the concepts of Digital Data Communication
2	**1**	Understand the basic concepts of Computer Networks
	2	Understand the Data Link Layer concepts
3	**1**	Comprehendthe addressing scheme in Computer Networks
	2	Understand the routing process in Computer Networks and Internet
4	**1**	Understand the process to process connection in network
	2	Understand the application layer protocols in Internet

SPECIFIC OUTCOMES:

MODULE – I: Physical Layer and Media

1.1 Understand the concepts of Digital Data Communication

1.1.1 Define Data

1.1.2 Define Signal

1.1.3 Differentiate between Analog and Digital Data

1.1.4 Describe the different Transmission Impairments in digital transmission

1.1.5 Explain the Digital and Analog transmission schemes – Digital to Digital, Digital to Analog, Analog to Digital and Analog to Analog

1.1.6 Analyse the Digital and Analog transmission schemes – Digital to Digital, Digital to Analog, Analog to Digital and Analog to Analog

1.1.7 List and Explain the guided transmission media

1.1.8 List and explain the unguided transmission media.

1.1.9 Differentiate between Guided and Unguided Transmission media.

MODULE – II: Data Link Layer

2.1 Understand the basic concepts of Computer Networks

2.1.1 Define Computer Networks

2.1.2 Classify the Computer Networks

2.1.3 Explain the Network software architectures

2.1.4 Explain ISO-OSI architecture

2.1.5 Explain TCP/IP Architecture

2.1.6 Compare Contrast the OSI and TCP/IP Architecture

2.2 Understand the Data Link Layer concepts

2.2.1 Explain the different types of Errors.

2.2.2 Differentiate between Error Detection and Error Correction

2.2.3 Explain Hamming code for error correction

2.2.4 Explain the Cyclic Redundancy Check code for error detection

2.2.5 Compute the Cyclic Redundancy Check code for error detection

2.2.6 Explain Framing

2.2.7 Compare the multiple access schemes Aloha, CSMA, CSMA/CD, CSMA/CA

2.2.8 Compare the schemes FDMA, TDMA and CDMA

2.2.9 Explain the Ethernet standard for Local Area Network

2.2.10 Draw and explain the Ethernet Frame format

2.2.11 Compare Fast and Gigabit Ethernet.

MODULE – III: Network Layer

3.1 Comprehend the addressing scheme in Computer Networks

3.1.1 Explain the logical addressing scheme

3.1.2 Explain the Classful addressing scheme of IPv4

3.1.3 Illustrate Classless addressing scheme of IPv4

3.1.4 Compare Classless and Classful addressing scheme of IPv4

3.1.5 Choose an appropriate addressing scheme and addresses for a given network

3.1.6 Draw and explain the IPv4 Datagram Structure

3.1.7 Explain fragmentation

3.1.8 Define Checksum

3.1.9 Describe the ICMP protocol- general format and types of ICMP messages

3.1.10 List the different categories of ICMP messages

3.2 Understand the routing process in Computer Networks and Internet

3.2.1 Define routing and routing protocol

3.2.2 Describe the routing protocol Distance vector routing protocol

3.2.3 Describe the routing protocol Link State routing protocol

3.2.4 Compare the two unicast routing protocols.

3.2.5 Explain IPv6 scheme of logical addressing scheme.

MODULE – IV: Transport and Application Layers

4.1 Understand the process to process connection in network

4.1.1 Define process to process delivery in networks

4.1.2 Define port address

4.1.3 Explain the UDP datagram format

4.1.4 Explain the checksum Calculation

4.1.5 Sketch and Explain the TCP Segment structure

4.1.6 Demonstrate the TCP Connection Establishment procedure

4.1.7 Demonstrate the TCP Connection Termination procedure

4.1.8 Distinguish between the TCP and UDP protocols in transport layer

4.1.9 Describe the flow Control mechanism of TCP

4.1.10 Describe the Congestion Control scheme of TCP

4.2 Understand the application layer protocols in Internet

4.2.1 Explain the DNS protocol

4.2.2 Sketch the DNS resolution process in the internet.

4.2.3 Describe the Email system

4.2.4 Describe the protocols FTP

4.2.5 Describe the application layer protocol HTTP.

CONTENT DETAILS

MODULE – I: Physical Layer and Media

Data and Signal - analog and Digital Data - Digital Signals - Transmission of Digital Signals - Transmission Impirements - Digital and Analog Transmission - Digital to Digital, Digital To Analog, Analog to Digital and Analog to Analog Communication- Guided and Unguided Transmission media.

MODULE – II: Data Link Layer

Computer Networks: Types of Computer Networks, Overview of ISO-OSI architecture- TCP/IP Architecture.

Data Link Layer: Types of Errors- Error Detection and Correction- Hamming Codes- Cyclic Redundancy Check- Checksum- Framing- Flow and Error Control- Multiple Access- Aloha, CSMA, CSMA/CD, CSMA/CA, FDMA, TDMA, CDMA, Ethernet- Standard Ethernet, Frame format, Fast and Gigabit Ethernet.

MODULE – III: Network Layer

Logical Addressing - IPv4 addressing- Classfull and Classless addressing- IPv4 Datagram Structure, Fragmentation and Checksum- ICMP- Unicast routing Protocols- Distance vector and Link State Routing- Introduction to IPv6 addressing.

MODULE – IV: Transport and Application Layers

Transport Layer: Process to Process Delivery, UDP- User Datagram, Checksum, TCP- Services, Segment, TCP connection, Flow Control, Congestion Control,

Application Layer: DNS and DNS Resolution, Email, FTP, HTTP

TEXT BOOK

1. Data Communication and Networking- Behrouz A Forouzan, 4th Eddition, Mc Graw Hill Publication

REFFERENCE

1. Computer Networks, Andrew S Tanenbaum, Pearson Education, 5th Edition.
2. TCP/IP Protocol Suite, Behrouz A. Forouzan, Fourth Edition, McGraw-Hill